Sports Medicine

Analysis of National Football League preseason injuries resulting in roster cuts: a retrospective review

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ABSTRACT

Background:
The National Football League (NFL) schedule consists of 16 regular season games and four additional preseason games each season. The structure of the NFL preseason allows franchises to evaluate 90 players in four games to determine which 53 should make the final roster. The purpose of this study was to investigate preseason injuries in NFL players and to identify risk factors for being cut from NFL teams after preseason injury.

Methods:
This study was a retrospective review of all injuries suffered by NFL athletes during the 2014 through 2019 NFL preseasons. The data were compiled using publicly available injury reports published on multiple websites.

Results:
We identified 110 injuries during the period of study involving players that were later cut from an NFL team during the preseason. The incidence of a preseason injury before being cut from the team was 1.59 per 1000 athletic exposures. The majority of NFL players who were cut after being injured were undrafted free agents (69 players, 63%). Rookies (38 players, 35%) most commonly were cut after being injured. Most athletes cut from NFL teams after preseason injury did not play another year in the NFL (82 players, 75%). The lower extremity was the most frequently injured body area, accounting for 45% (49 players) of all injuries.

Conclusions:
No prior NFL experience and being undrafted in the NFL draft were two common characteristics identified among athletes that sustained preseason injury and were later cut from the roster.

Level of Evidence:
Level III.

Key Words
NFL, preseason, injury, roster cut

INTRODUCTION

The National Football League (NFL) schedule consists of 16 regular season games and four additional preseason games each season. During the regular season NFL teams are allowed to roster only 53 players. However, during the preseason NFL teams can roster up to 90 players until they determine who ultimately will make the team.[1] The involvement of the league's top players in preseason games has been diminishing; thus, preseason games often consist of fringe players competing for a roster spot.[2] The stakes are high for fringe players who may or may not make the team, as NFL salaries are guaranteed only upon making the final roster.

Two provisions exist for players who are injured in the preseason if the team wishes to keep them on the roster for the regular season: the Physically Unable to Perform (PUP) list and the Injured Reserve (IR) list. Depending on contract structure, these typically allow the player to be paid despite being unable to play. This structure incentivizes NFL teams to use injury exceptions only on their best players because each injured player's salary detracts from the team's overall salary cap, despite the fact that they are not contributing during games, and this can inhibit teams from trading and acquiring players

Preliminary data from the NFL suggests concussions are increasing during the preseason. During the 2017 preseason, an increase in the occurrence of concussions was recorded from 26 in 2016 to 45 the following year.[3] The 2018 preseason had the highest concussion rate since 2015.[4] Football Outsiders,[5] a statistical group specializing in NFL analysis, reports that there are more than 300 injuries during training camp each season. These injuries lead to a 'survival of the fittest' phenomenon where injury-prone players are cut from teams. As such, players injured in the preseason may forgo substantial income as NFL contracts are largely distributed on a per game basis.[5]

The structure of the NFL preseason allows franchises to evaluate 90 players in four, arguably, meaningless games to determine which 53 should make the final roster. This system inherently exposes borderline recruits to the risk of injury and subsequently

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being cut from the team. The purpose of this study was to investigate preseason injuries in NFL players and to identify common characteristics of players cut from NFL teams following preseason injury. Our hypothesis was that players drafted in later rounds (eg, rounds six or seven) more likely will be cut after an injury when compared to players drafted in the first two rounds.

MATERIALS AND METHODS

Ethical Review and Study Design

This study was a retrospective review of all injuries suffered by NFL athletes during the 2014 through 2019 NFL preseasons. The data was compiled using publicly available injury reports published on multiple websites.[3,6] Each injury was corroborated with two additional publicly available online sources, which reported on the same data collected from the original source, in accordance with previously published protocols.[7–13] Institutional review board approval was not necessary or obtained for this study because all data included in the study were publicly available.

Data Collection

Reportable injuries were based on the criteria that the injury necessitated a team physician referral or emergent care and resulted in game time being missed. Athletes injured during the preseason who were then subsequently cut from the team were included in the study. Team records, available on each individual team website, were used to collect demographic information. Our compiled data included player's name, position, team, pathology, date of injury, round drafted, years in the NFL when cut, if the injury required surgery, whether or not the athlete made another 53 man roster, years active in the NFL after injury, and date of being cut from the team. Finally, preseason injuries that did not result in a player being released from the team were excluded from this study. The sample size was determined by the inclusion of all injuries suffered by NFL athletes during the 2014 through 2019 NFL preseasons.

Statistical Analysis

An analysis was performed on the collected NFL data from the 2014 through the 2019 NFL preseasons. The mean, frequency, and incidence rate of injury were calculated using SPSS 25 (SPSS Inc, Chicago, Illinois). For data analysis, an athletic exposure (AE) was defined as one athlete appearing in one game. No distinction was made between athletes with regard to playing time. Game injury rates were calculated per 1000 AE. The maximum AE during a single game was 180, in which all 90 players from each team participated in a contest. We did not calculate the incidence rate of injury for practices due to a lack of standardized protocol and unreliable injury reporting methods.[14] Injury incidence rate per 1000 AE was calculated using the following formula: (Total number of injuries x 1000)/Total number of AE. Total number of AE was calculated using the following data: 90 Players on each roster × 32 Teams × 4 Preseason games × 6 seasons = 69,120 AE.

RESULTS

There were 32 teams competing in the NFL from the 2014 through the 2019 regular seasons (Table 1). Each NFL team plays four games during the preseason per year. NFL rosters are limited to 53 players during the season. Each NFL team must eliminate 37 roster spots prior to starting the regular season. Therefore, a minimum of 1,184 players are cut each year during the NFL preseason. After being cut from one NFL team, a player may sign with another NFL team if offered a contract. We identified 110 injuries during the period of study involving players who later were cut from an NFL team during the preseason. The incidence of a preseason injury prior to being cut from the team was 1.59 per 1000 AE. The most common position cut after an injury was defensive back (24 players, 22%), followed by linebacker (19 players, 17%), and tight end (16 players, 15%) (Table 2).

TABLE 1. Number of players cut per team

Team	Number of players (%)
49ers	3 (4)
Bears	4 (4)
Broncos	3 (3)
Browns	5 (5)
Bucs	4 (4)
Cardinals	5 (5)
Chargers	5 (5)
Chiefs	2 (2)
Colts	7 (7)
Cowboys	4 (4)
Dolphins	6 (5)
Eagles	6 (5)
Falcons	4 (4)
Giants	6 (5)
Jaguars	4 (4)
Jets	3 (3)
Lions	2 (2)
Panthers	8 (7)
Patriots	3 (3)
Rams	4 (4)
Ravens	1 (1)
Redskins	3 (3)
Saints	2 (2)
Seahawks	5 (5)
Steelers	7 (6)
Texans	2 (2)
Titans	1 (1)
Vikings	1 (1)
Total	110

TABLE 2. Player positions

Position played	Number of players (%)
OL	11 (10)
Cornerback	8 (7)
DB	24 (22)
DE	5 (5)
DT	3 (3)
FB	1 (1)
LB	19 (17)
QB	1 (1)
RB	8 (7)
TE	16 (15)
WR	14 (13)
Total	110

DB, defensive back; DE, defensive end; DT, defensive tackle; FB, fullback; LB, linebacker; OL, offensive lineman; QB, quarterback; RB, running back; TE, tight end; WR, wide receiver.

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TABLE 3. Round of draft pick

Round drafted	Number of players (%)
1	1 (1)
2	1 (1)
3	9 (8)
4	5 (5)
5	7 (6)
6	4 (4)
7	14 (13)
Undrafted	69 (63)
Total	110

Most NFL players who were cut after being injured were undrafted free agents (69 players, 63%); players drafted in the seventh round (14 players, 13%) were the second most common, followed by players drafted in the third round (nine players, 8%) (Table 3). Similarly, with regard to NFL experience, rookies (38 players, 35%) most commonly were cut after being injured, followed by players with 1 year of NFL experience (24 players, 22%), and players with 2 years of NFL experience (18 players, 16%) (Table 4). The numbers of players cut per year are listed in Table 5. Most athletes who were cut from NFL teams secondary to preseason injury did not play another year in the NFL (82 players, 75%) (Table 6). Approximately 25% of the dataset (28 players) made another NFL team's roster after being cut (Table 7). Only three (3%) of the 110 reported injuries had surgery for management of their injury; however, these data were often unavailable and were likely underreported (Table 8).

The lower extremity was the most frequently injured body area, accounting for 45% (49 injuries) of all injuries (Table 9). The lower extremity injury incidence of 0.70 per 1000 AE was significantly higher than any other body area. The next most common injured body area was 'undisclosed', which accounted for 30% (33 injuries) of all injuries followed by upper extremity injuries at 10% (11 injuries).

Analysis of pathology demonstrated similar results; ankle sprains were noted to be the most common injury (14 injuries, 13%) (Table 10). The second most common was knee injury (12 injuries, 11%); however specific details regarding knee pathology were not available. Shoulder injuries (nine injuries, 8%) were the

TABLE 4. Years in the National Football League (NFL) when cut

Years in the NFL when cut	Number of players (%)
0	38 (35%)
1	24 (22)
2	18 (16)
3	15 (14)
4	6 (5)
5	3 (3)
6	4 (4)
7	0 (0)
8	0 (0)
9	0 (0)
10	1 (1)
11	1 (1)
Total	110

TABLE 5. Year of cut

Year	Number of players (%)
2019	29 (26)
2018	14 (13)
2017	35 (32)
2016	16 (15)
2015	2 (2)
2014	14 (13)
Total	110

TABLE 6. Years in the NFL after injury

Years in the NFL after injury	Number of players (%)
0	82 (75)
1	19 (17)
2	5 (5)
3	2 (2)
4	1 (1)
5	0 (0)
6	1 (1)
Total	110

TABLE 7. Players who made another 53-man roster

Did the player make another 53-man roster after being cut from team?	Number of players (%)
Yes	28 (25)
No	82 (75)
Total	110

TABLE 8. Injured players managed surgically

Surgery	Number of players (%)
Yes	3 (3%)
No	107 (97%)
Total	110

TABLE 9. Injury rate by body area

Body area	Number of players (%)
Lower extremity	49 (45)
Upper extremity	11 (10)
Torso	4 (4)
Head	6 (5)
Spine/back	7 (6)
Undisclosed	33 (30)
Total	110

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TABLE 10. Injury rate by pathology

Pathology	Number of players (%)
Achilles tendon rupture	2 (2)
Ankle sprain	14 (13)
Back injury	6 (5)
Calf	1 (1)
Rib fracture	4 (4)
Collarbone fracture	1 (1)
Concussion	5 (5)
Metacarpal fracture	1 (1)
Foot fracture	4 (4)
Groin strain	2 (2)
Hamstring	8 (7)
Head	1 (1)
Hip contusion	2 (2)
Knee	12 (11)
Leg	3 (3)
Quadriceps tendon rupture	1 (1)
Neck	1 (1)
Shoulder	9 (8)
Undisclosed	33 (30)
Total	110

third most common injury observed, but again specific details about the shoulder pathology were not available.

DISCUSSION

The risk of injury associated with football is significant, and all-cause injury rates have been reported as high as 65 per 1000 AEs in NFL preseason games.[1,15] We reported an incidence of preseason injury prior to being cut from an NFL team of 1.59 per 1000 AE. Thus, only about 1 in 40 preseason injuries result in a player being cut from an NFL team. Unfortunately, we were unable to identify any literature reporting all-cause injury rates of career-ending injuries in NFL players; however, Mai *et al.*[16] demonstrated in a case series that anterior cruciate ligament (ACL) reconstruction and patellar tendon repair led to significantly shorter NFL careers after surgery.

Our results, which demonstrated knee pathology as the most common injury pattern in players who were later cut from an NFL team, support previously published NFL preseason injury demographic data that found knee sprains to be the most common NFL preseason injury.[1,15] NFL athletes requiring ACL reconstruction have been shown to experience significant declines in statistical performance up to 3 yr after injury.[16,17] The findings from our dataset show that lower extremity injuries were the most common type of preseason injury athletes suffered prior to being cut. Furthermore, our data indicated that the most common position cut following a preseason injury was defensive back (24 players, 22%). These findings support previously published NFL injury data that demonstrated that defensive backs have a significantly higher odds ratio of suffering hamstring and ankle injuries.[18] Mai *et al.*[16] also reported similar findings that defensive backs compromised the largest proportion of injuries in their dataset.[16]

To our knowledge, this is the first study to report on injuries prior to preseason roster cuts among NFL athletes. This study analyzed preseason cuts, demographic data, player outcomes after being cut, and overall risk factors leading to player cuts after preseason injuries. The purpose of this investigation was to identify injury patterns and risk factors among NFL athletes who were cut from the team roster after experiencing preseason injuries. This study provides important insight into the implications for NFL athletes sustaining preseason injuries, including possible career-ending roster cuts.

Because of limited roster space and team financial constraints associated with placing players on an IR list, injuries during the preseason can be especially deleterious on the career of an NFL athlete. The most common characteristic identified in this study of players who were cut from the team after a preseason injury was no NFL experience prior to the injury. Rookie athletes cut after a preseason injury constituted 75% of the athletes identified in this study. Thus, no prior NFL experience was identified as a key attribute among players who were cut after sustaining a preseason injury. This finding is likely multifactorial; Chambless *et al.*[19] demonstrated a higher injury rate in rookie professional athletes, which they attributed to inadequate adjustment to the higher level of competition. Furthermore, NFL teams are typically less financially invested in rookie players, specifically seventh round draft picks or undrafted free agents because of their contract structure. Rookies who are drafted in late rounds receive relatively small signing bonuses compared to their veteran counterparts. Because contracts are paid out on a per-game basis during the regular season, NFL teams are only required to pay the cut player their signing bonus. There is very little financial incentive to keep injured rookie players on the roster as demonstrated by our findings that less than 13% of players cut after preseason injuries had 4 or more years of NFL experience prior to injury of interest. Another common characteristic among athletes who were cut after preseason injuries was being undrafted from the NFL Draft. Most athletes (69 athletes, 63%) who were identified in our cohort and were cut after preseason injuries were undrafted, and 13% (14 players) were drafted in the seventh round. Together, players undrafted or those drafted in the seventh round constituted 76% of all NFL players who were injured and later cut in the preseason from 2014 to 2019. Only two (2%) of players who were cut after preseason injuries were drafted in the first or second round.

Injuries can mark the end of an athlete's career. Among NFL players from 2014 to 2019 who were injured during the preseason, and later cut, only 25% (28 players) made a 53-man NFL roster in the future. Thus, a preseason injury preceding a roster cut marked the end of an athlete's career in an overwhelming majority of cases.

Limitations and Future Perspectives

Our study had limitations that should be taken into consideration when interpreting the results. This was a retrospective review, and thus was subject to the intrinsic constraints of retrograde study. Publicly available information regarding NFL injuries lacks specific pathology details, and the treatments received for those injuries, such as surgical intervention. Moreover, specific reasons (ie, player talent, team finances, injury) for player cuts are not made available to the public. Therefore, we cannot objectively conclude injury resulted in athletes being cut, and conclusions can only be drawn from available information. An NFL injury database, NFL Injury Surveillance System (NFLISS), does exist, and we

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encourage those with access to examine injury-specific data, surgical interventions, and postinjury outcomes for NFL players who experienced preseason injuries prior to a preseason roster cut that year.

CONCLUSIONS

The current study reported an incidence of preseason injury prior to a preseason roster cut from an NFL team of 1.59 per 1000 AE. Thus, approximately 1 in 40 preseason injuries result in a player being cut from an NFL roster. The most common position cut after an injury was defensive back (24 players, 22%), followed by linebacker (19 players, 17%), and tight end (16 players, 15%). Most of the NFL players who were cut after being injured were undrafted free agents (69 players, 63%) or players drafted in the seventh round (14 players, 12%), accounting for 75% of roster cuts after injury during the study period. Knee pathology was the most common injury pattern in athletes cut from an NFL team after a preseason injury; however, ankle sprains were the most common injury. Most of the athletes who were cut from NFL teams secondary to preseason injury did not play another year in the NFL. Of the players who were cut, 25% (28 players) were picked up by another NFL team during the preseason. We conclude that draft position (seventh round or undrafted) and knee pathology are risk factors for preseason roster cuts NFL players.

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